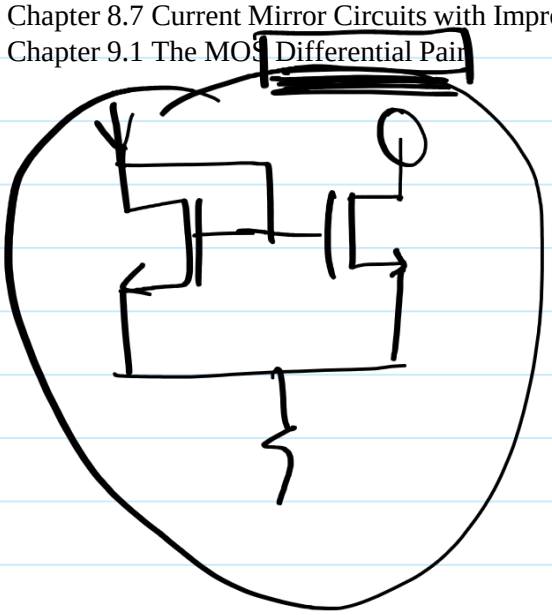
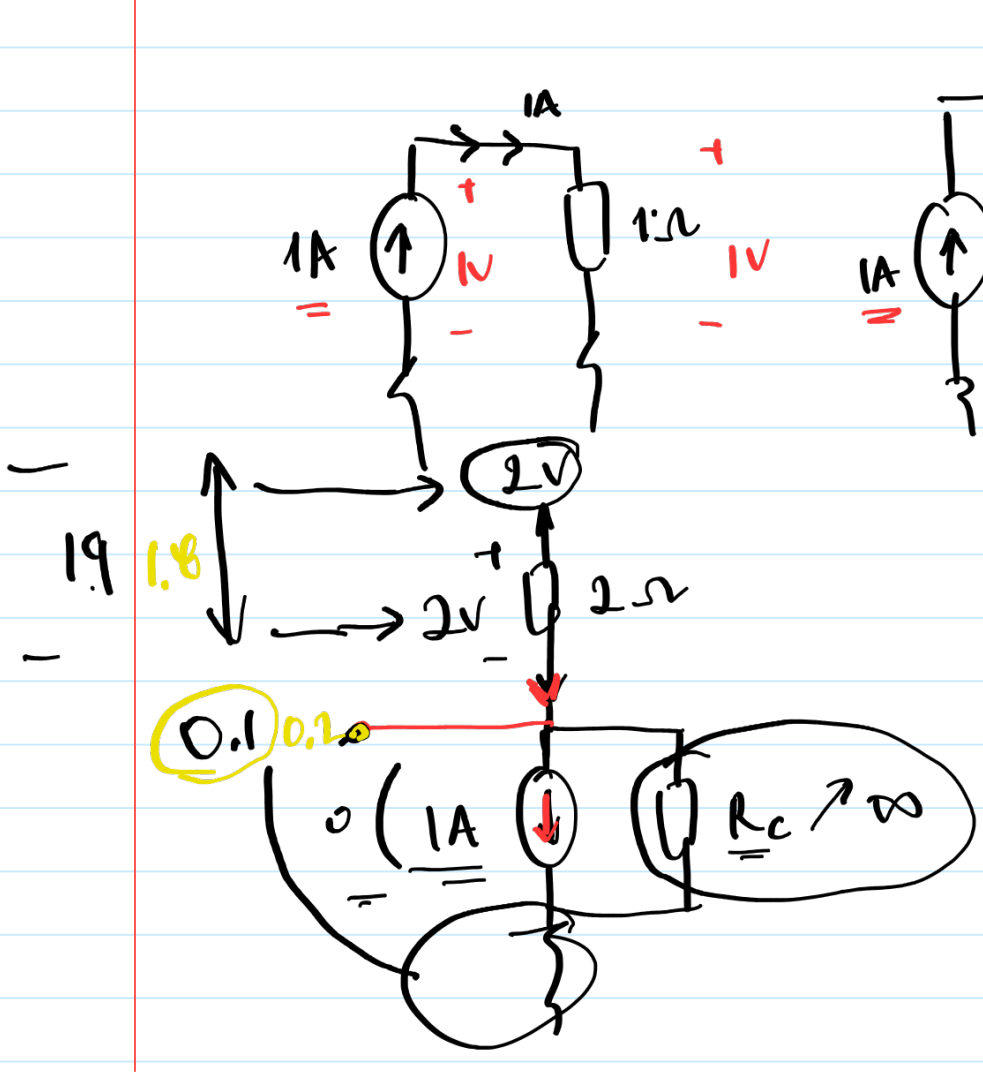
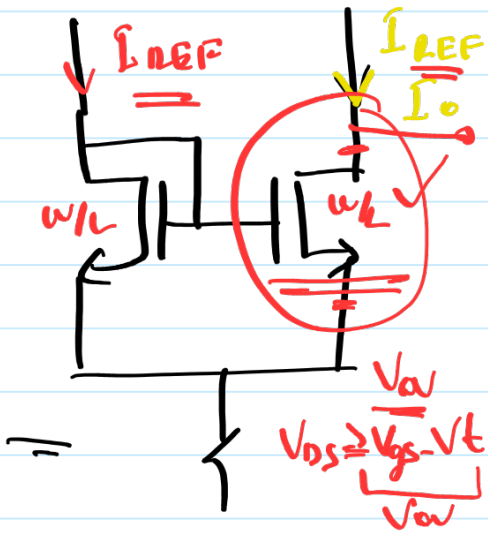
Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

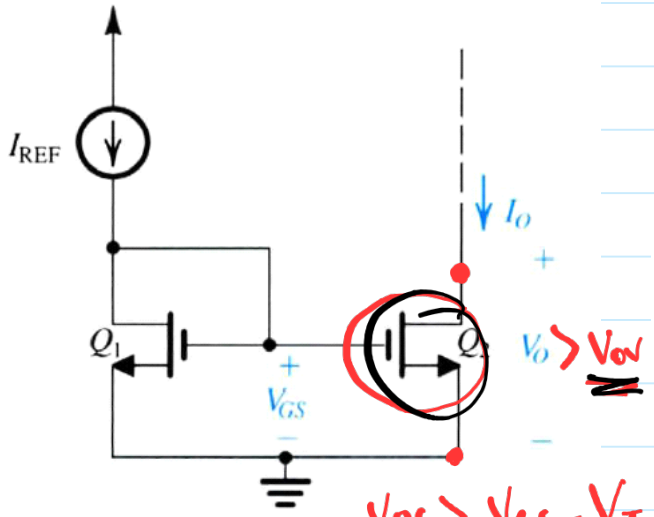
- Chapter 8.7 Current Mirror Circuits with Improved Performance
- Chapter 9.1 The MOS Differential Pair







Recap



$$I_O|_{\text{nominal}} = \frac{(W/L)_2}{(W/L)_1} I_{REF}$$

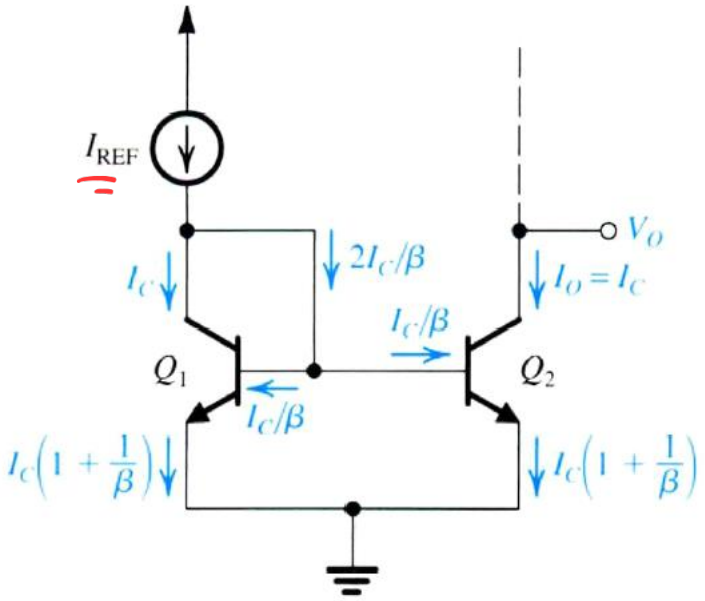
$$I_O = \frac{(W/L)_2}{(W/L)_1} I_{REF}\left[1 + \frac{V_O - V_{GS}}{V_{A2}}\right]$$

$$R_o = r_{o2}$$

$V_O > V_{ov}$

$V_{OS} \geq V_{GS} - V_T = V_{ov}$

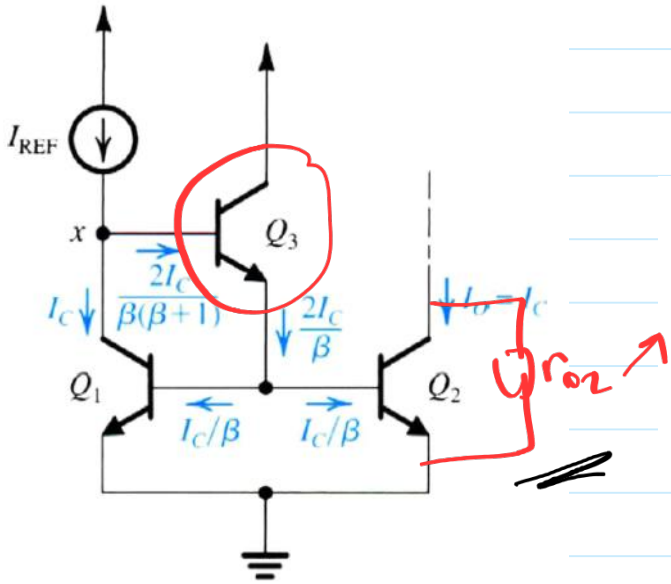
0.2 0.3



$$\frac{I_O}{I_{REF}} = \frac{I_C}{I_C\left(1+\frac{2}{\beta}\right)} = \frac{1}{1+\frac{2}{\beta}}$$

$$\frac{I_O}{I_{REF}} = \frac{m}{1+\frac{m+1}{\beta}}$$

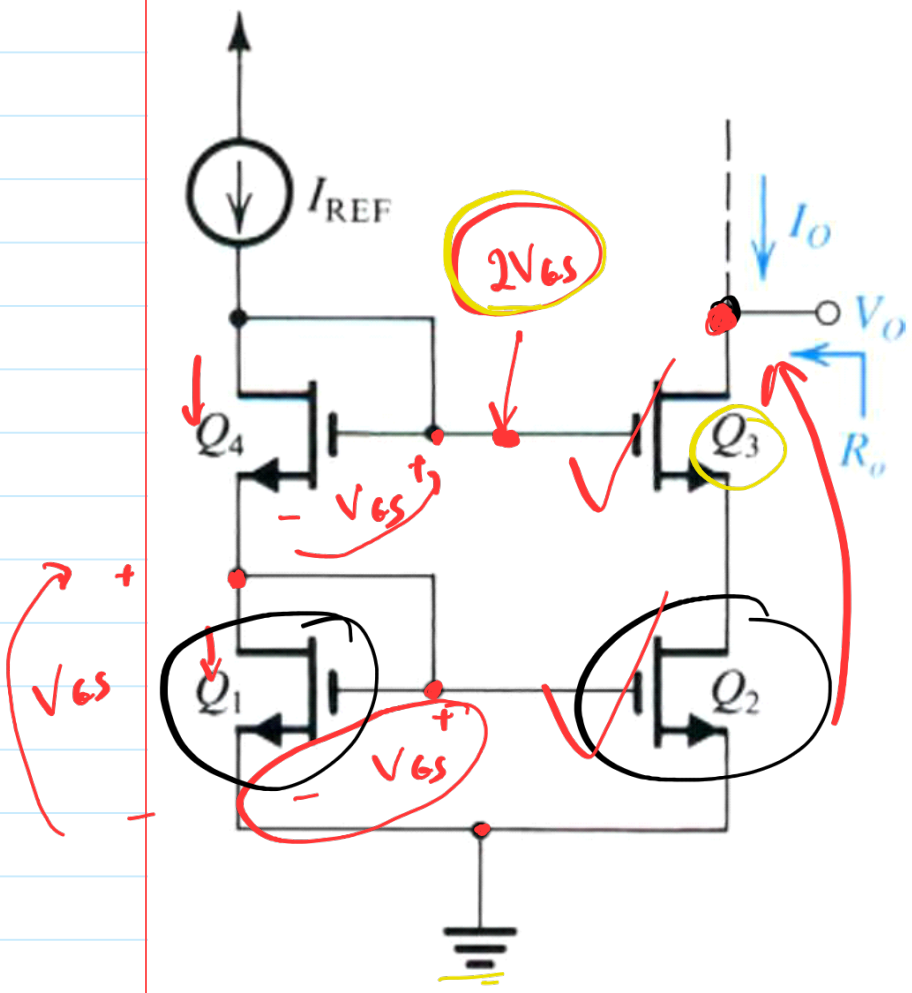
$$I_O = I_{REF}\frac{m}{1+\frac{m+1}{\beta}}\left(1+\frac{V_O - V_{BE}}{V_{A2}}\right)$$



r_{o2} ↗

$$\frac{I_O}{I_{REF}} = \frac{1}{1+2/(\beta^2+\beta)} \simeq \frac{1}{1+2/\beta^2}$$

The Cascode Current Mirror



$R_o = r_o(g_{m3}r_{o3})$

$V_{DS_3} \geq V_{GS_3} - V_T$

$V_D - V_S \geq V_G - V_S - V_T$

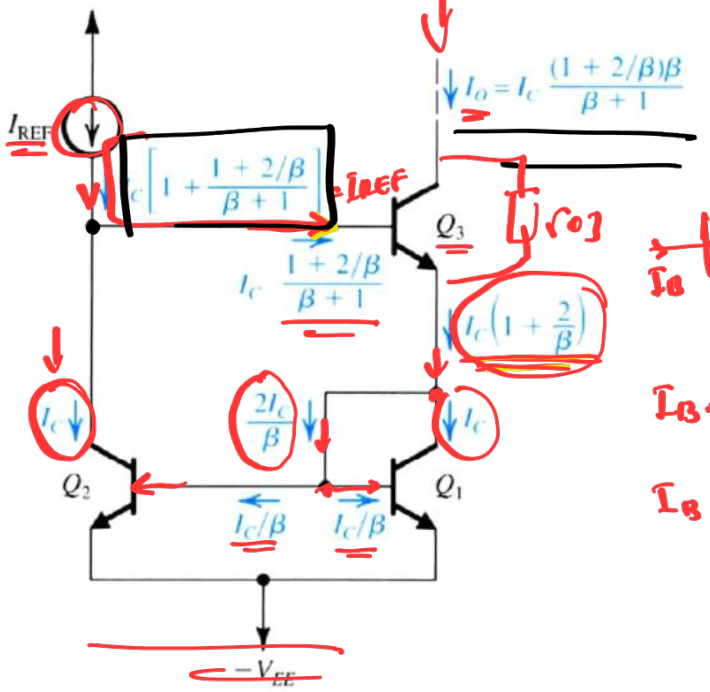
$V_{D_3} \geq V_{G_3} - V_T$

$V_{D3} \geq 2V_{GS1} - V_T$

$V_{D3} \geq 2(V_{GS1} - V_T) + V_T$

$V_{D3} \geq 2V_{OV} + V_T$

The Wilson BJT Current Mirror



$R_O = \frac{1}{2} r_{O3} \cdot \beta$

$I_O = I_C \frac{(1+2/\beta)\beta}{\beta+1}$

$I_C = \alpha I_E = \frac{\beta}{\beta+1} I_E$

I_B

I_E.

$I_B = \frac{I_C}{\beta+1}$

$I_B = \frac{I_C}{\beta}$

$I_C = I_S \cdot e^{V_{BE}/V_T}$

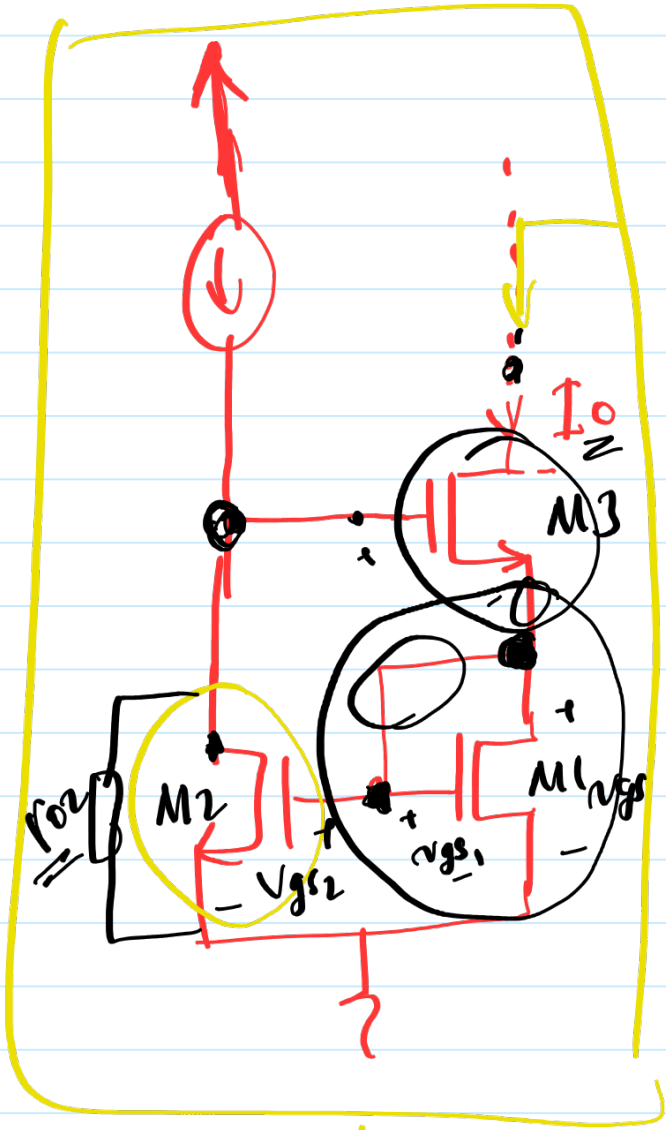
$$\frac{I_O}{I_{REF}} = \frac{I_C\left(1+\frac{2}{\beta}\right)\beta(\beta+1)}{I_C\left[1+\left(1+\frac{2}{\beta}\right)/(\beta+1)\right]}$$

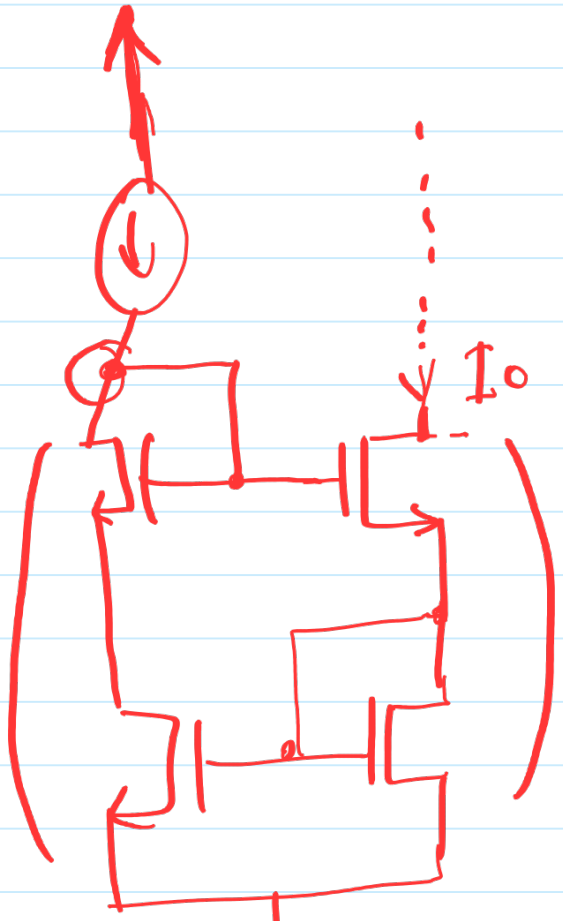
$$\frac{I_O}{I_{REF}} = \frac{1}{1+\frac{2}{\beta(\beta+2)}} \cong \frac{1}{1+2/\beta^2}$$

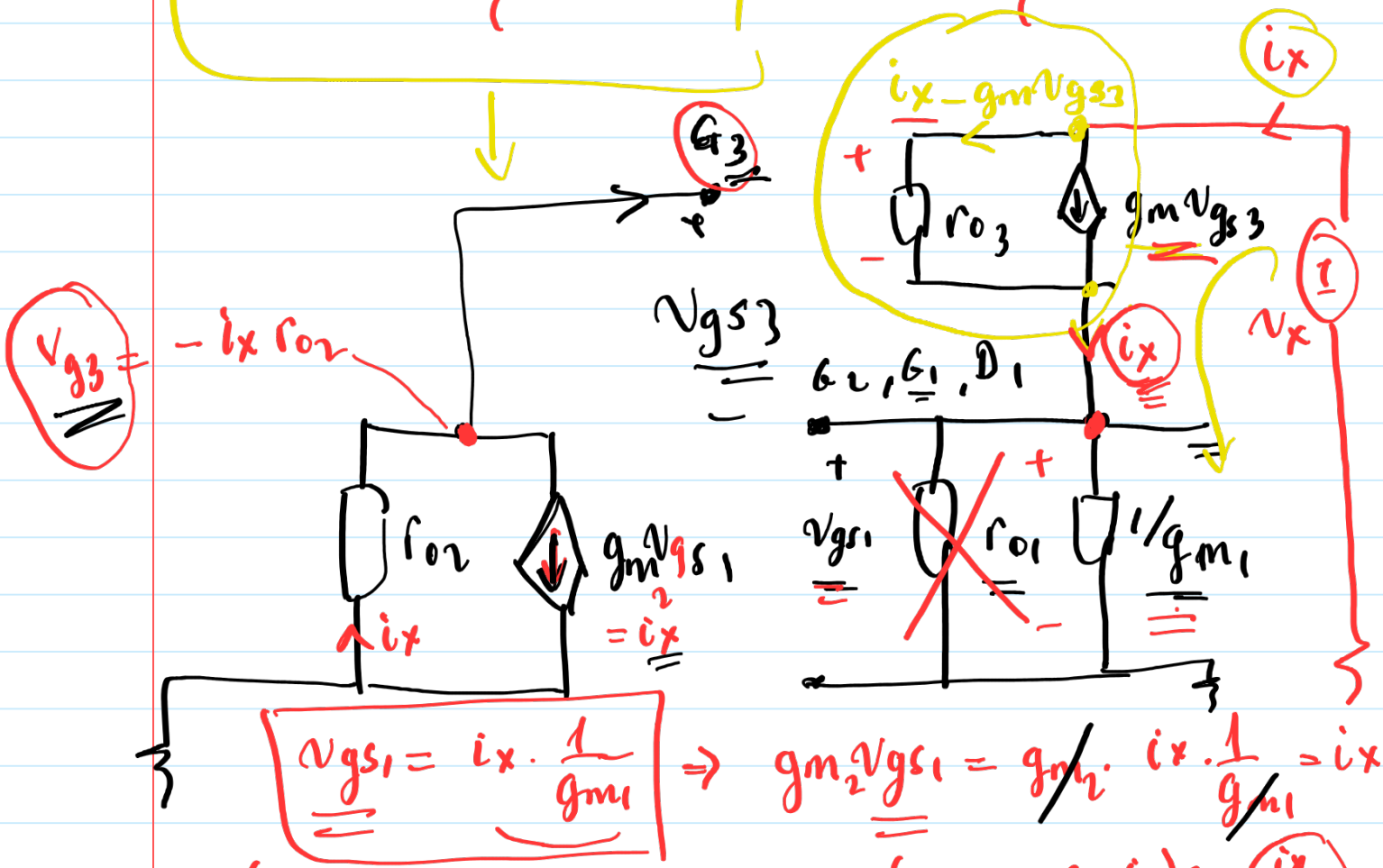
$\beta \to 100$

$$R_O = \frac{1}{2}\beta_3 r_{O3}$$

The Wilson MOS Mirror



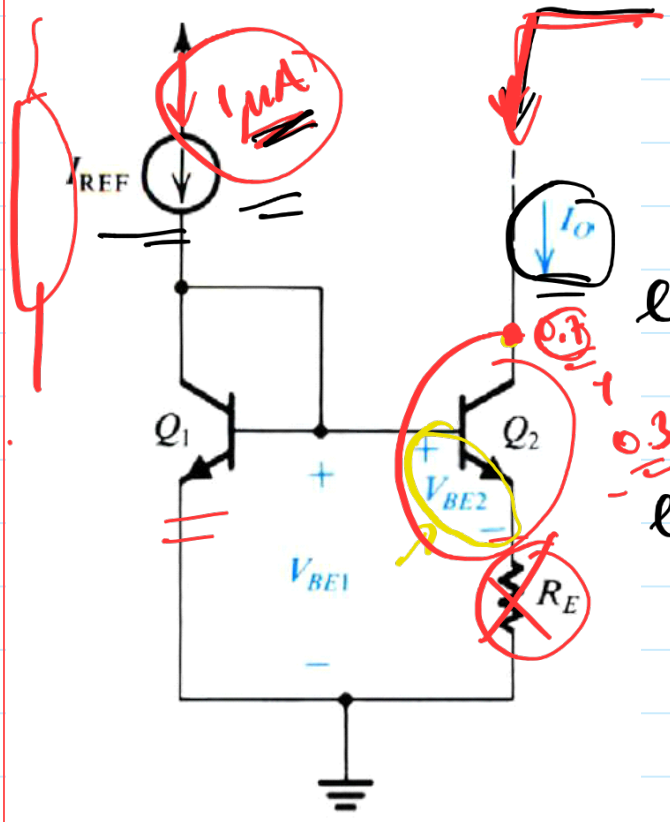




$$v_{gs1} = i_x \cdot \frac{1}{g_{m1}} \Rightarrow g_{m2} v_{gs1} = g_{m2} \cdot i_x \cdot \frac{1}{g_{m1}} = i_x$$

$$v_x = (i_x + g_{m3} r_{o2} i_x) r_{o3} \approx \frac{i_x}{g_{m1}} \Leftarrow \text{Ao} \approx (g_{m3} r_{o3}) r_{o2}$$

The Widlar Current Source



$$R_{out} = \left[1 + g_m (R_E \,||\, r_\pi)\right] r_o$$

$$I_E = I_S \, e^{V_{BE}/V_T}$$

$$\ln\left(\frac{I_E}{I_S}\right) = \ln\left(e^{V_{BE}/V_T}\right)$$

$$\ln \frac{I_E}{I_S} \cdot V_T = V_{BE}$$

$$V_{BE1} = V_T \ln\left(\frac{I_{REF}}{I_S}\right)$$

$$V_{BE2} = V_T \ln\left(\frac{I_O}{I_S}\right)$$

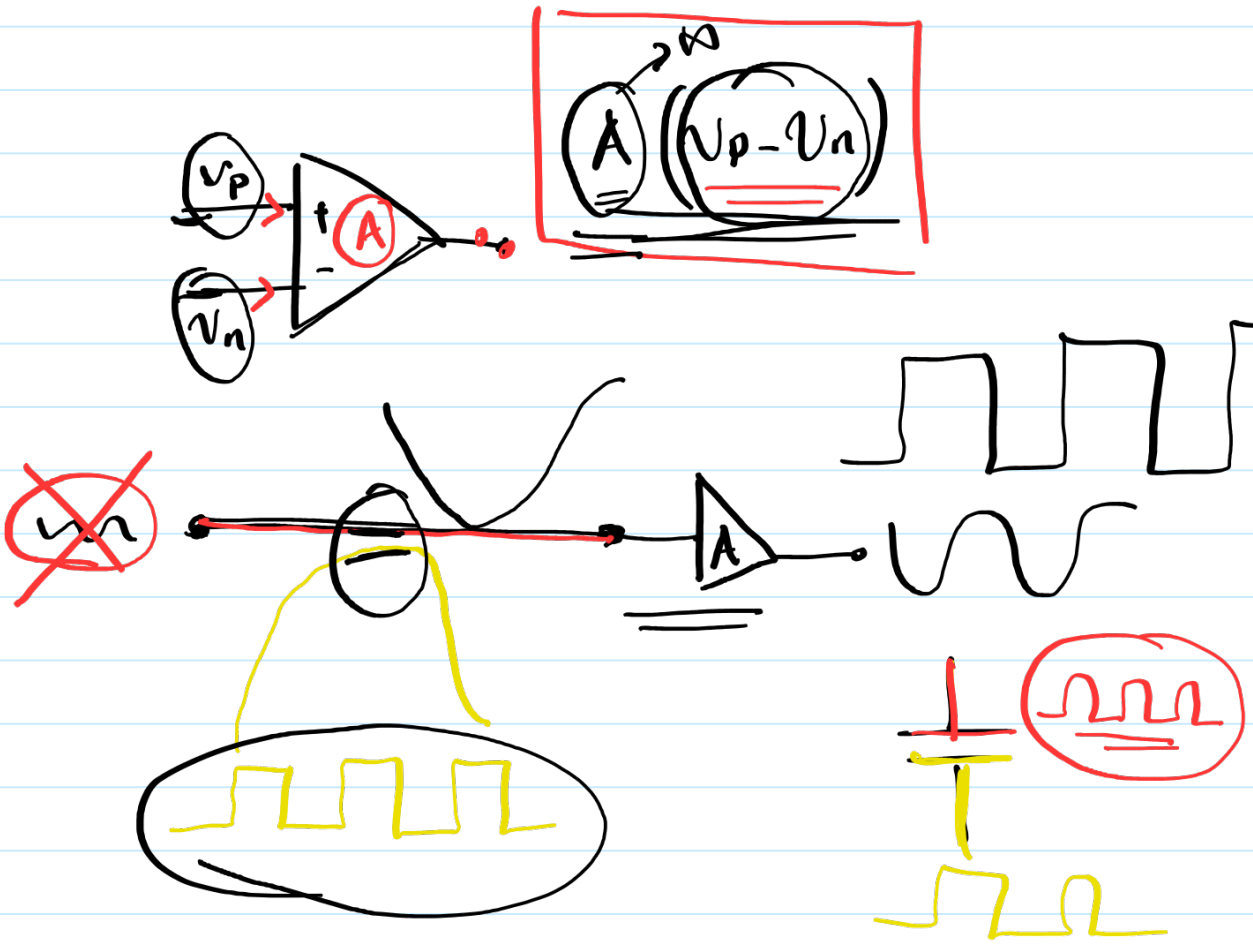
$$V_{BE1} - V_{BE2} = V_T \ln\left(\frac{I_{REF}}{I_S} \cdot \frac{I_S}{I_O}\right)$$

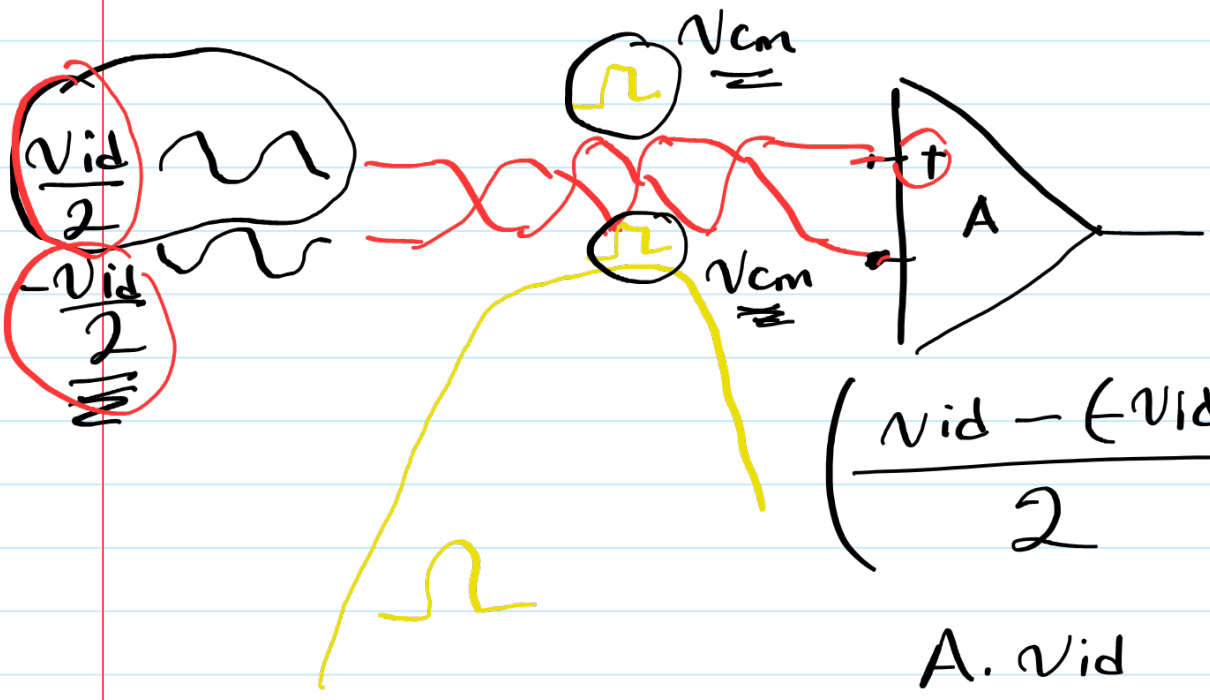
$$V_{BE1} - V_{BE2} = V_T \cdot \ln\left(\frac{I_{REF}}{I_O}\right)$$

$$V_{BE1} = V_{BE2} + I_O \cdot R_E$$

$$I_O R_E = V_T \ln\left(\frac{I_{REF}}{I_O}\right)$$

MOS Differential Pair with [redacted]put





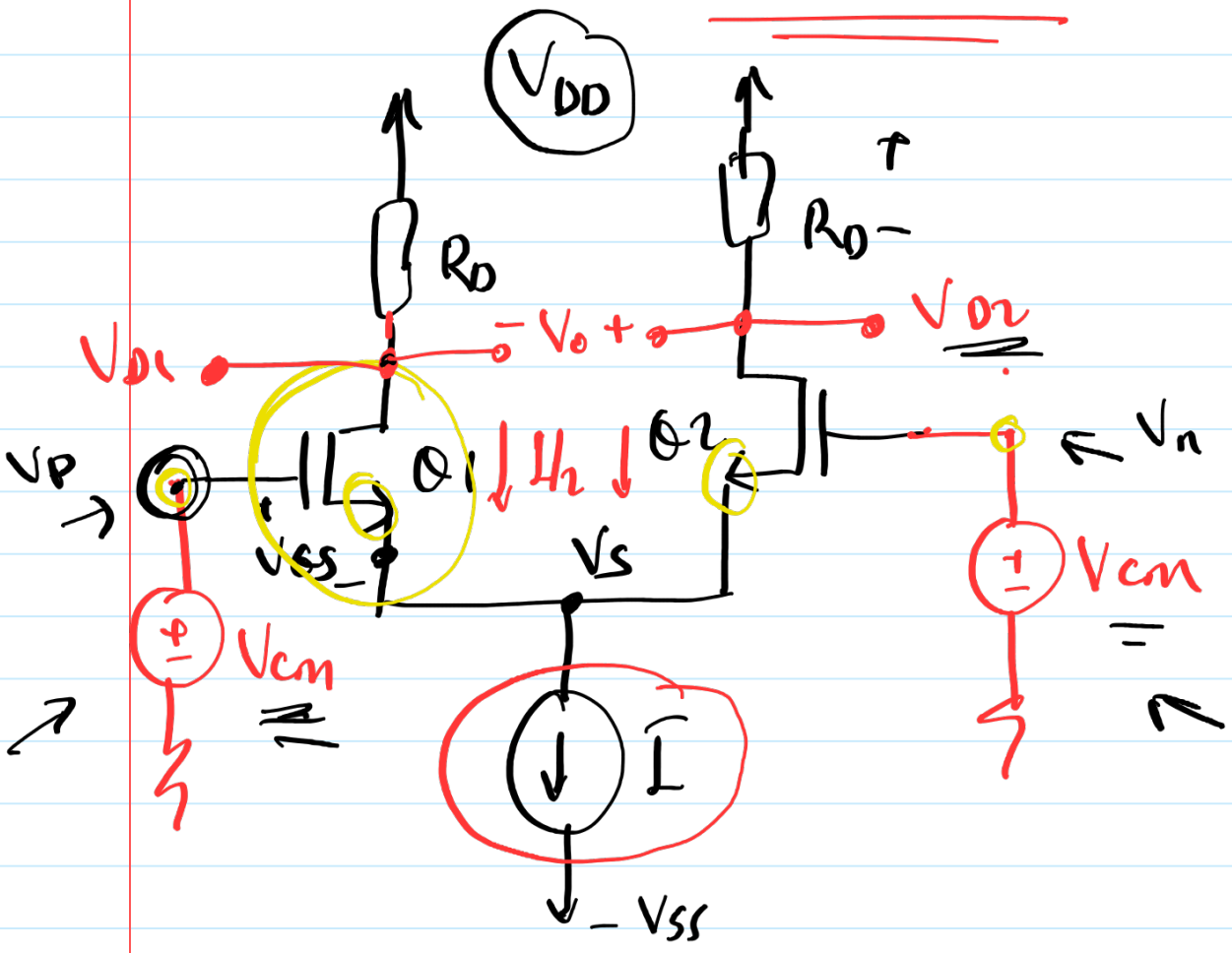
$$\left(\frac{v_{id} - (-v_{id})}{2}\right) A =$$

$$A \cdot v_{id}$$

$$v_p = \frac{v_{id}}{2} + v_{cm} \qquad v_n = -\frac{v_{id}}{2} + v_{cm}$$

$$A(v_p - v_n) = \frac{v_{id}}{2} + v_{cm} + \frac{v_{id}}{2} - v_{cm} = v_{id}$$

MOS Differential Pair with Common Input



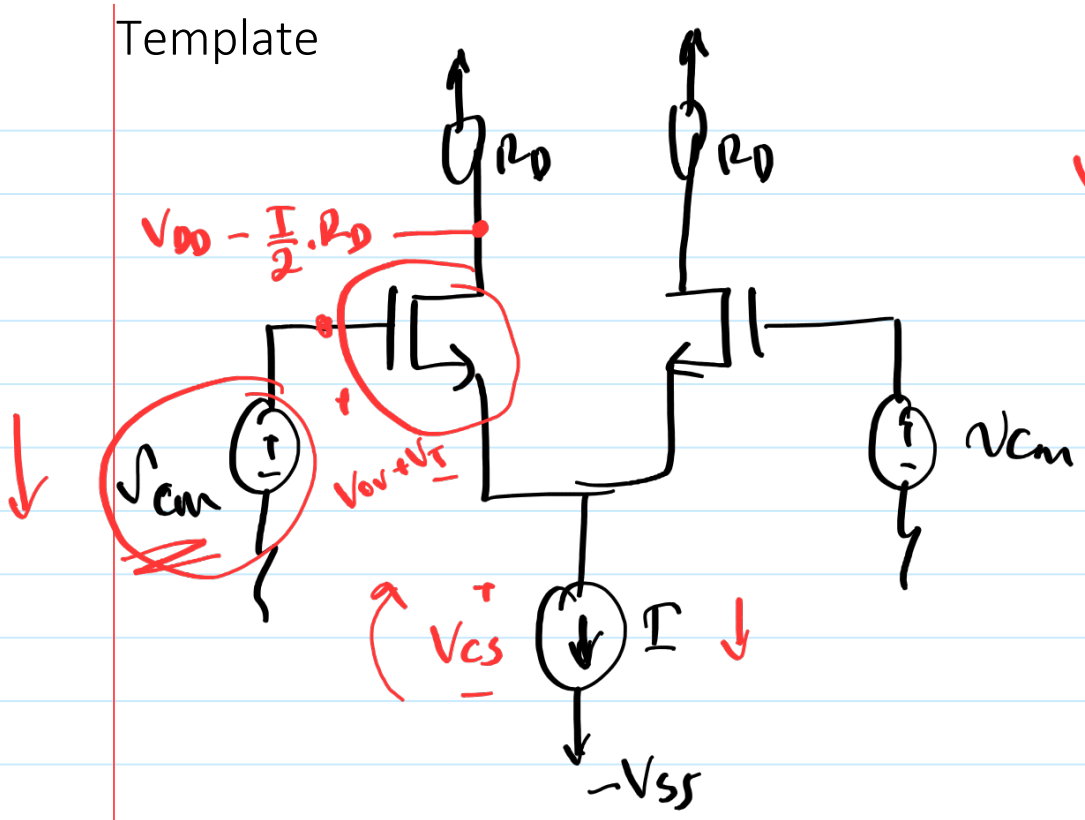
$$V_S = V_{CM} - V_{GS}$$

$$\frac{I}{2} = I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2$$

$$V_{ov}^2$$

$$V_{ov} = \sqrt{\frac{I}{k_n' W/L}}$$

$$V_{D2} = V_{D1} = V_{DD} - \frac{I}{2} R_D \qquad V_o = V_{D2} - V_{D1} = 0$$



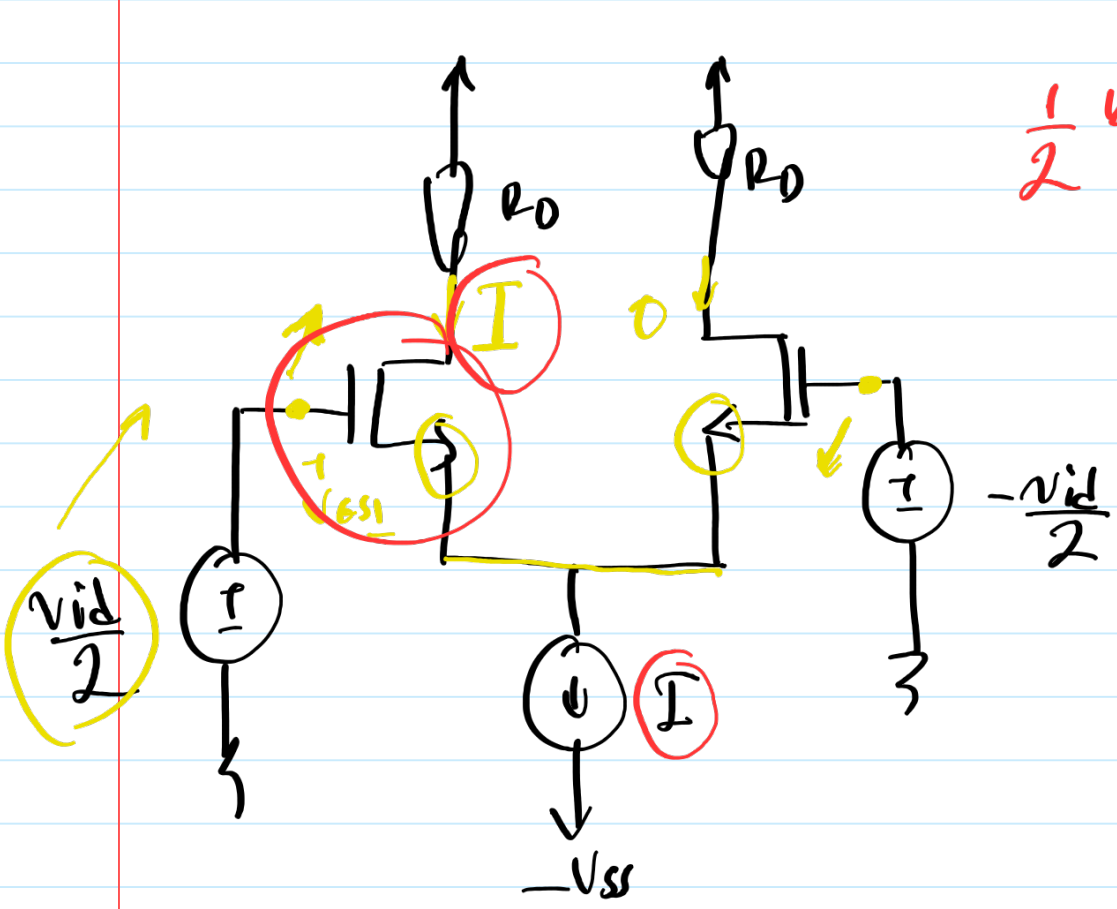
$V_{GS} = V_T + V_{OV}$

$V_O > V_G - V_T$

$V_{DD} - \frac{I}{2} R_D > V_{cm} - V_T \Rightarrow V_{DD} - \frac{I}{2} R_D + V_T > V_{cmax}$

$-V_{SS} + V_{CS} + V_{OV} + V_T \leq V_{cmin}$

~~Template~~ Differential Circuit with V_{id}



$\frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2$

$$I_D = \frac{1}{2} \left(k_n' \frac{W}{L} \right) (V_{GS1} - V_t)^2$$

$$V_{GS1} = \sqrt{\frac{2I}{k_n' W/L}} + V_t = \sqrt{2}\, V_{ov}' + V_t$$

Large-Signal Operation

Large-Signal Operation (cont.)

Template